GLASGOW COLLEGE UESTC

Main Paper - Fall 2023-2024

REAL TIME COMPUTING & ARCH (UESTC4019)

Date: 2nd January 2024
Time: 09:30 – 11:30 (2 hours)

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.
Show all work on the answer sheet.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

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Question 1

(a) How a real-time embedded system differs from a conventional embedded system. [2]

(b) (i) Why are floating-point numbers needed to represent very large and very small numbers in computer and embedded systems? [3]

(ii) Represent the floating number -121.312500 in IEEE-754 single precision standard floating point. Show all the calculations and steps to receive full marks. [8]

(c) In computer organization, performance refers to the speed and efficiency at which a computer system can execute tasks and process data. The following table shows the execution times, in seconds, for two different benchmark programs on three different machines A, B, and C.

Benchmark	**Processor**		
	A (sec)	**B (sec)**	**C (sec)**
1	20	10	40
2	40	80	20

Table Q1(c)

(i) Suggest a matrix that can be used to measure and compare these computer systems' performance. Justify your selection. [4]

(ii) Compute the arithmetic means value for each system using first A as the reference machine and then using B as the reference machine. Which machine is the slowest based on these two calculations? Based on the calculations and results found, comment if this approach is realistic or misleading. [2+1+1]

(iii) Now, compute the geometric means value for each system using first A as the reference machine and then using B as the reference machine. Which machine is the slowest based on these two calculations? Based on the calculations and results found, comment if this approach is realistic or misleading. [2+1+1]

Continued overleaf

Question 2

(a) List and briefly define three techniques commonly used for performing I/O. [2+2+2=6]

(b) Consider a computer system that contains an I/O module controlling a simple keyboard/printer teletype. The following registers are contained in the processor and connected directly to the system bus.

INPR: Input Register, 8 bits
OUTR: Output Register, 8 bits
FGI: Input Flag, 1 bit
FGO: Output Flag, 1 bit
IEN: Interrupt Enable, 1 bit

Keystroke input from the teletype and printer output to the teletype are controlled by the I/O module. The teletype is able to encode an alphanumeric symbol into an 8-bit word and decode an 8-bit word into an alphanumeric symbol.

(i) Explain clearly how the processor, using the first four registers listed in this problem, can achieve I/O with the teletype. [4]

(ii) Suggest another approach which can be adopted to make the above function more efficient. [4]

(c) A two-way set-associative cache has lines of 16 bytes and a total size of 8 kB. The 64-MB main memory is byte addressable. Calculate and show the configuration of designed main memory addresses. [6]

(d) A particular system is controlled by an operator through commands entered from a keyboard. The average number of commands entered in an 8-hour interval is 60.

(i) In the current design, the processor scans the keyboard every 100 ms. How many times will the keyboard be checked in an 8-hour period? [2]

(ii) As an embedded system engineer, do you think the approach used in part d(i) is the optimum method for this system? If your answer is Yes, justify. If your answer is No, propose a better approach (s). [3]

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Question 3

(a) The operation of the processor is determined by the instructions it executes, referred to as machine instructions or computer instructions.

(i) There are four common elements of machine instruction. List any three of these elements. [3]

(ii) Assume computing the Y= C + (D×E). Write programs to execute Y using typical "one-address instructions" architecture. [4]

(iii) In addition to bitwise logical operations, most machines provide a variety of shifting and rotating functions, namely logical shift, arithmetic shift, and rotate. Assume "10100110" as an input data. What are the results for each operation below? [4]

- Logical right shift (3 bits)
- Logical left shift (3 bits)
- Arithmetic right shift (3 bits)
- Arithmetic left shift (3 bits)

(iv) Figure Q3(a) schematically indicates two different addressing modes (see plots (A) and (B)). What is the name of each addressing mode, the principal advantages and disadvantages of each mode separately? [6]

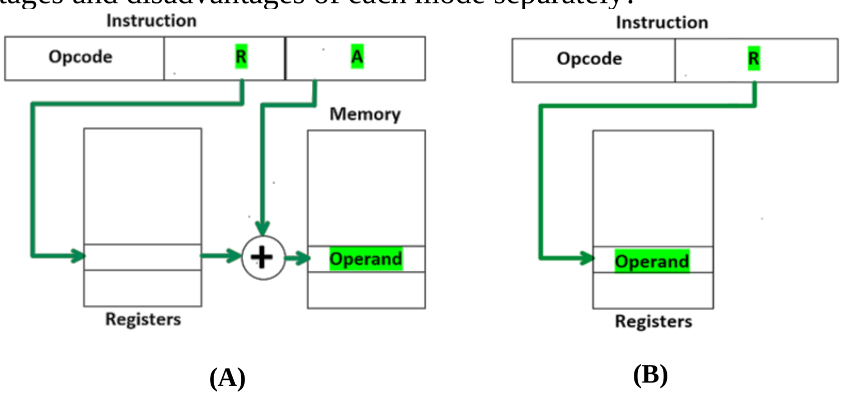

Figure Q3(a)

(b) In the processor organization, some main requirements are placed (the things that the processor must do). Mention four of these requirements. [4]

(c) Considering Instruction-Level Parallelism, and Superscalar Processors, answer the following questions:

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(i) What fundamental limitation to parallelism is placed in the assembly code below: [1]

```
1:  ADD R1, R2, R3
2:  SUB R1, R4, R5
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(ii) How the code above can be modified to eliminate the limitation? [1]

(iii) In general terms, we can group superscalar instruction issue policies into three categories. List any two of these categories. [2]

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Question 4

(a) Considering the conceptual processing of a superscalar. Select any three processes (stages) and explain an overview of program execution in each relevant stage. [6]

(b) There are four common categories in grouping systems with parallel processing capability. List any three of these categories. [3]

(c) You are working on a computation that can be divided into two parts: a parallelizable portion and a sequential portion. The parallelizable portion accounts for 60% of the original execution time, and the remaining 40% is sequential and cannot be parallelized. You currently have 6 parallel processors to speed up the computation.

(i) Calculate the theoretical speedup you can achieve with the current setup of 6 parallel processors. [2]

(ii) How many additional processors would you need to add to achieve a 50% increase in speedup compared to the current setup (with 6 parallel processors)? [2]

(d) Consider a set of processes arriving at a computer system with the following burst time requirements. Assume that Non-Preemptive Shortest-Job-First (SJF) scheduling is used.

Process	Arrival Time	Burst Time
P1	0	6
P2	1	4
P3	2	8
P4	3	3

(i) Calculate the completion order of these processes by providing a Gantt chart to illustrate the execution sequence. [8]

(ii) Calculate the average turnaround time for all processes. [2]

(iii) Calculate the average waiting time for all processes. [2]

End of question paper